

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 16

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Single: $5.8 \times 20 \times 12 = 1392 \text{ W}$ • Argon: $1.6 \times 20 \times 12 = 384 \text{ W}$ • Reduction = 1008 W (accept 1010)	3
01.2	• A wider gap of trapped gas reduces conduction because gas is a poor conductor • If the gap is too wide convection currents start in the gas, carrying heat across; the optimum depends on the gas and window size	2
01.3	• Low-e coating reflects long-wavelength infrared back into the room, reducing radiation losses • Argon / krypton / xenon have lower thermal conductivity than air so less conduction across the gap	2
01.4	• Panes must be held apart by small spacers which cannot be long, so the gap is only about 0.2 mm • Used where there is not room for thick gas-filled multiple glazing (eg historic buildings)	2
01.5	• Each extra layer saves less energy (diminishing returns) while cost rises, so beyond the optimum the cost of insulation exceeds the value of energy saved	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-231; AQA spec 3.3.4.2.2.	
02.1	• The amount of energy contained in a given mass of the resource (or harnessed per kg of equipment for renewables) • High energy density means smaller quantities are needed so storage and transport are easier and cheaper (eg 75 t of aviation fuel carries a 400 t aircraft 5600 km) • High energy density fuels reach high temperatures needed for smelting / steam turbines	3

Question	Marking guidance	Marks
02.2	• $500\,000 / 28 = 17\,857\text{ kg}$ • $= 17.9\text{ tonnes}$ (accept 18; book quotes 13 t for 0.7% enriched fuel)	2
02.3	• The electricity they generate can power high-energy-requirement technologies • Electricity can be converted to high energy density secondary fuels, eg hydrogen by electrolysis, or stored (batteries, pumped storage)	2
02.4	• Intermittent: not available all the time, eg tidal (but the times are predictable so alternatives can be planned) • Unpredictable: intermittent and the timing / amount cannot be forecast accurately, eg wind, solar	2
02.5	• Gaseous natural gas has a very low energy density by volume; LNG at $-160\text{ }^{\circ}\text{C}$ has 160 times the energy density so a ship can carry a worthwhile amount	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-178; AQA spec 3.3.2.	
03.1	• There is no gaseous / atmospheric reservoir of phosphorus (no equivalent of nitrogen fixation) • Most phosphorus compounds have low solubility so little is available in soil water • Phosphorus lost to marine sediments only returns through very slow processes: mountain building (uplift) and weathering of rocks	3
03.2	• Fungal hyphae form a symbiotic relationship with plant roots and spread through a large volume of soil • Greatly increasing the surface area for absorbing (poorly mobile) phosphate ions, passed to the plant in exchange for carbohydrates	2
03.3	• Mining of calcium phosphate rock (and past mining of guano deposits, eg Peru, Nauru, now largely exhausted) • Treated to make more soluble ammonium phosphate fertilisers, mobilising phosphorus into the environment • Fertiliser runoff / eroded soil / sewage effluent carry phosphates into water bodies causing eutrophication (with nitrates); harvesting removes phosphorus from farmland	3
03.4	<i>Any 2 from:</i> • Use biological wastes (manure, sewage sludge, compost) as fertiliser to return phosphates rather than mining rock • Breed crops that absorb phosphate more efficiently • Maintain conditions for mycorrhizal fungi (less tillage, fewer fungicides, organic matter) • Recover phosphate from sewage effluent by iron(III) sulfate precipitation; reduce soil erosion	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 156-157; AQA spec 3.2.4.4.	
04.1	• About 40-50 traps • Because the running mean stops changing / fluctuating with fewer than this the mean is unreliable	2
04.2	• Widely scattered results (large standard deviation) mean a single sample may not be representative • More samples are needed so that the effect of variability is averaged out and the mean is representative / significance can be assessed	2

Question	Marking guidance	Marks
04.3	<i>Any 3 from:</i> • Same type and size of pitfall trap, set for the same length of time • Traps emptied at the same time of day / same weather conditions in both habitats • Same number of traps and same method of positioning (random or systematic) in each habitat • Other variables that could affect results (eg rainfall, temperature) monitored or controlled	3
04.4	• Flying insect activity changes with weather over minutes to weeks (temperature, wind) and diurnally • Samples on different occasions are needed to produce a mean that represents the typical value or to identify trends • The interval between samples should match the rate at which the factor changes, found from a preliminary study	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
05.1	• Numbers fell from about 210 pairs in 1945 to a minimum of about 25 in 1965 (steepest fall 1955-1960) • Numbers then recovered / rose steadily after 1970, reaching about 190 by 1985 (not yet back to 1945 level)	2
05.2	• $(200 - 25) / 200 \times 100$ • = 87.5%	2
05.3	<i>Any 4 from:</i> • DDT is persistent (remains for decades) and liposoluble so it bioaccumulates in prey (small birds) and biomagnifies in sparrowhawks (top predator) • Sub-lethal doses caused thin eggshells / reduced breeding success so the population declined even without adult deaths • Higher doses caused sterility / death • After restrictions, environmental concentrations slowly fell (persistence explains the delay before recovery) • Breeding success recovered so the population rose	4
05.4	<i>Any 2 from:</i> • DDT persists in soil / sediments / food chains for many years so exposure continued • Birds of prey are k-selected: few young per year, slow to mature so populations grow slowly • Other pressures (habitat loss, persecution)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
06.1	• Mutagen: an agent that changes the chemical structure / base sequence of DNA (causes mutations), eg ionising radiation, UV, PCBs, cadmium, asbestos • Carcinogen: a mutagen that causes cancer / uncontrolled cell division producing a tumour • Teratogen: an agent that causes birth abnormalities by preventing normal gene expression (inhibiting proteins / enzymes) without changing DNA structure, eg mercury, 2,4,5-T	3

Question	Marking guidance	Marks
06.2	• Teratogens do not change the DNA structure, so the abnormality is not in the genes and cannot be passed on • Mutagens change DNA; if the mutation is in a gamete / egg / sperm (gonadic effect) it can be inherited	2
06.3	• Somatic: mutation in a general body cell, affecting the individual only (eg cancer / tumour) • Gonadic: mutation in cells of the ovaries or testes / gametes / embryo, which may cause abnormalities in offspring	2
06.4	• Two or more pollutants whose effects interact to produce a different (usually more serious) effect • The pollutants themselves do not react to form a new pollutant (that would be a secondary pollutant) • Example: ozone damages leaf cuticles so sulfur dioxide can cause more damage to the exposed cells; smoke and sulfur dioxide in smogs; neonicotinoids with fungicides	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 239-240; AQA spec 3.4.1.	
07.1	• Natural pressure of gas above the oil (or dissolved in it) or water beneath forces oil up the production well; pump-jacks ('nodding donkeys') increase flow • Pressure falls as oil is removed so the remaining oil no longer flows to the well	2
07.2	• Water or natural gas is pumped down an injection well to maintain the pressure and flow of oil (recovery to about 40%) • Captured CO₂ can be pumped down instead, increasing oil recovery while storing the CO₂ underground	2
07.3	<i>Any 3 from:</i> • Steam injection (from combustion or solar parabolic concentrators) to heat the oil and reduce its viscosity • Controlled underground combustion to heat the oil • Detergents / solvents to reduce surface tension so oil flows more easily • Bacteria that partially digest heavy oil into lighter oils and produce CO₂ to maintain pressure	3
07.4	• Many wells can be drilled from a single platform, reducing land take / number of rigs • Deposits under locations where rigs cannot be placed (urban areas, sensitive habitats) can be reached • Drilling can follow softer strata (quicker) and target multiple small reservoirs up to 10 km away, increasing total recovery	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 187-188; AQA spec 3.3.4.1.	
08.1	• When there is a difference in level between the sea and the basin: on the ebb (about 6-8 hours) water flows out through the turbines • Also possible on the flood when the sea is higher than the basin; no generation when levels are equal (slack water)	2
08.2	• Two periods of slack water in each 12.5-hour cycle (twice in about 25 hours) give no flow; the timing does not match demand • Tides depend on the positions of the Moon and Sun which are known, so output can be forecast years ahead and alternative supplies planned	2

Question	Marking guidance	Marks
08.3	• Spring tides: Sun and Moon aligned so their gravitational effects combine giving a large tidal range; neap: effects oppose giving a small range • Energy depends on the mass of water and height difference, so output is much greater at springs	2
08.4	• When demand is low, surplus electricity pumps water into the lagoon above sea level • The stored water is released through the turbines later when demand is high (pumped storage)	2
08.5	• Large tidal range (eg Severn Estuary) • Estuaries, bays or headlands / channels that focus the flow and increase velocity	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 214-215; AQA spec 3.3.4.1.	
09.1	• The void fills with water naturally; wetlands are uncommon / valuable habitats in most regions • The new habitat may have greater wildlife value than the farmland lost; low cost compared with filling and levelling	2
09.2	• Housing: ground must be stable (no subsidence), no toxic metal wastes (otherwise industrial use is safer) • Farmland: landscape not too steep or uneven, no toxic materials, sufficiently fertile soil (topsoil replaced)	2
09.3	• Species that cannot be caught / found, or whose habitat requirements are unknown or hard to recreate, are lost; transplanted plants often fail	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 140, 142; AQA spec 3.2.3.6.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Stages of scientific method illustrated by ozone: identification (Antarctic UV data), hypothesis (Rowland-Molina 1974), data collection / analysis (Dobson units, satellites, balloons confirming CIO), proposed solutions, enactment (Vienna 1985, Montreal 1987) • Evidence of success: ozone hole area and depth stabilised and beginning to shrink; CFC concentrations in the atmosphere falling; UV levels not rising further; global ozone loss limited to about 4% • Reasons for success: international recognition of serious consequences (skin cancer); agreement of nearly every country; legally binding targets with a fund for developing countries; relatively few producers / products; alternatives (HCFCs, HFCs, pump sprays, hydrocarbons) already available and cheap; no lifestyle change needed • Limits: persistence of CFCs means recovery takes decades; some ODSs still used; HFC replacements are greenhouse gases • Climate change contrast: greenhouse gases come from almost every human activity (energy, transport, agriculture, industry) in every country; fossil fuels underpin economies so control requires major lifestyle and economic change • Agreements (Kyoto, Paris) have weaker / non-binding targets, disputes between developed and developing nations; emissions still rising; natural fluctuations, time delays and tipping points make cause-effect less clear to the public • Judgement: ozone control was a technical substitution problem; climate change is a systemic one, so the ozone success is only partly transferable	9
10.2	• Vienna Convention for the Protection of the Ozone Layer - no legally binding goals	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-113, 105-106; AQA spec 3.2.1.3, 3.2.1.2.	

Question	Marking guidance	Marks
11.1	Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Features of fertile soil: water, nutrients, air, organic matter, pH, biota, texture, structure, depth; slow rate of soil formation vs rapid erosion (dynamic equilibrium broken) • Causes of degradation: vegetation removal, ploughing vulnerable soils, overgrazing, reduced organic matter and biota, cultivating steep slopes, compaction, salinisation from irrigation, nutrient depletion, pollution • Types of erosion: rain splash, runoff, wind, slumping; consequences: reduced productivity, river and reservoir sedimentation, flooding, coastal sedimentation, particulates, desertification, landslides • USLE as a tool for estimating loss and testing management • Management: long-term crops, low / zero tillage, contour ploughing, tied ridging, terracing, stone rows, windbreaks, multicropping / strip cropping, mulching, organic matter, livestock control, drainage / pH / nutrient management, conservation of soil biota • Links to sustainable agriculture, carbon storage, hydrological cycle, circular economy • Evaluation and judgement on which strategies are most effective in different environments Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169, 328-337; AQA spec 3.2.5, 3.5.1.3, 3.5.1.5.	25

Question	Marking guidance	Marks
11.2	Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Nature of the problem: interconnected atmosphere-hydrosphere-biosphere systems; changes over different time and spatial scales; natural fluctuations (solar output, orbit) masking or exaggerating anthropogenic change; time delay between cause and effect (ocean heat capacity) • Limitations of historic data: sparse global coverage, unsophisticated equipment, heat island bias, no atmospheric composition records • Proxy data and their reliability: dendrochronology, coral growth rings, pollen in dated sediments, ice cores (gas bubbles, isotope ratios, volcanic ash layers, compaction problems) • Modern monitoring: long-term data sets (Mauna Loa), electronic equipment, satellites (LEO vs geostationary), Argo floats, remote sensing of ice thickness / area / sea level • Feedback mechanisms and tipping points adding uncertainty; ecological uncertainty (species survival, distribution, fragmentation) • Computer models: how they are tested and refined; importance of accurate representative data; scenario-based predictions and ranges • Uncertainty over ENSO / Conveyor / jet stream responses • Evaluation and conclusion: confidence in the overall trend and human cause despite detailed uncertainty; role of peer review and international bodies (IPCC) Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90, 100-104, 10; AQA spec 3.2.1.2, 3.1.1.3.	25